STATISTICS FOR ECONOMISTS · LESE152

Practice Set 2

STANDARD / 실전 | 원본 PS1과 유사한 풀이량 및 문항 난도

Name: _____ **Student ID:** _____

Exam structure: 10 MCQ + 10 T/F + 3 short-answer questions Total: 100 points | Suggested time: 80-95 min

Instructions: Choose one answer per MCQ. Mark True/False and provide a brief explanation. Show all major calculation steps in Part C. Check units, conditioning events, and independence assumptions.

Standard normal: $\Phi(0.5)=0.6915$, $\Phi(1)=0.8413$, $\Phi(1.2816)=0.9000$, $\Phi(1.5)=0.9332$, $\Phi(1.645)=0.9500$, $\Phi(1.96)=0.9750$, $\Phi(2)=0.9772$.

For the standard normal, $\Phi(-z)=1-\Phi(z)$. Useful exponentials: $e^{-1}=0.3679$, $e^{-2}=0.1353$, $e^{-3}=0.0498$, $e^{-4}=0.0183$.

Percentile position convention: $Lp=(p/100)(n+1)$, with linear interpolation between neighboring sorted observations.

PART A · MULTIPLE CHOICE (10 × 2 = 20 points)

MC1. Annual output of 100 farms tracked over five years forms which data structure?

(a) Cross-section (b) Time series
(c) Panel (d) Census

MC2. Chebyshev's inequality guarantees at least what fraction within 2 standard deviations?

(a) 50% (b) 68%
(c) 75% (d) 95%

MC3. A and B are independent, P(A)=0.5 and P(B)=0.4. Find P(A ∪ B).

(a) 0.20 (b) 0.50
(c) 0.70 (d) 0.90

MC4. A depot receives Poisson(3) trucks per hour. Find P(exactly one truck).

(a) 0.0498 (b) 0.1494
(c) 0.2240 (d) 0.9502

MC5. T ~ Uniform(12, 36) minutes. Find P(T < 20).

(a) 1/4 (b) 1/3
(c) 1/2 (d) 2/3

MC6. For sorted data 10,20,30,40,50,60,70, what is the 75th percentile using Lp=(p/100)(n+1)?

(a) 50 (b) 55
(c) 60 (d) 65

MC7. X ~ Binomial(10, 0.2). What is E(X)?

(a) 0.2 (b) 1

(c) 2 (d) 8

MC8. A sample correlation of -0.9 indicates:

(a) Strong negative linear association (b) Strong positive linear association

(c) Causality from X to Y (d) Independence

MC9. T ~ Normal(80, 5^2). Find P(T > 90).

(a) 0.0228 (b) 0.1587

(c) 0.5000 (d) 0.9772

MC10. Revenue R and cost C are independent with sd(R)=3 and sd(C)=4. Find sd(R-C).

(a) 1 (b) 5

(c) 7 (d) 25

PART B · TRUE / FALSE (10 × 3 = 30 points)

TF1. Under random sampling with finite mean, the sample mean is an unbiased estimator of the population mean.

Mark: True [ ] False [ ] Explanation: _____

TF2. If two events are mutually exclusive and both have positive probabilities, they must be independent.

Mark: True [ ] False [ ] Explanation: _____

TF3. P(A | B) is always the same as P(B | A) whenever both are defined.

Mark: True [ ] False [ ] Explanation: _____

TF4. If E(X) = 4 and $E(X^2) = 21$, then Var(X) = 5.

Mark: True [ ] False [ ] Explanation: _____

TF5. Adding the same positive constant to every observation leaves the coefficient of variation unchanged.

Mark: True [ ] False [ ] Explanation: _____

TF6. If X and Y are independent, E(XY) = E(X)E(Y).

Mark: True [ ] False [ ] Explanation: _____

TF7. A continuous PDF f(x) can exceed one at some points.

Mark: True [ ] False [ ] Explanation: _____

TF8. A Poisson random variable can take negative integer values.

Mark: True [ ] False [ ] Explanation: _____

TF9. An exponential random variable has the memoryless property.

Mark: True [ ] False [ ] Explanation: _____

TF10. For X ~ Binomial(n,p) with 0<p<1, Var(X)=np.

Mark: True [ ] False [ ] Explanation: _____

PART C · SHORT ANSWERS (50 points)

C1. Weekly market sales

Weekly produce sales (in crates) over 8 observations: 15, 18, 20, 22, 25, 27, 30, 43.

(a) [5 pt] Compute the mean and median and comment on the distribution.

(b) [5 pt] Compute Q1, Q3, and the upper outlier fence using (n+1) interpolation.

(c) [5 pt] Compute the sample standard deviation and coefficient of variation.

C2. Two crop-pest tests

A pest affects 12% of fields. Conditional on infestation status, two tests are independent. Each test has sensitivity P(+ | D)=0.85 and false-positive rate P(+ | D^c)=0.10.

(a) [5 pt] Calculate the probability that the first test is positive.

(b) [5 pt] Given one positive test, find P(D | +).

(c) [5 pt] Given two positive tests, find P(D | +,+). State where conditional independence is used.

C3. Delivery orders, profit, and preparation

Orders per day X: x = 0, 1, 2, 3; P(X=x) = 0.15, 0.35, 0.35, 0.15.

(a) [5 pt] Calculate E(X), Var(X), and sd(X).

(b) [5 pt] Profit Y = 30X - 12 (thousand won). Calculate E(Y) and sd(Y).

(c) [5 pt] Four bills are paid independently; each is late with probability 0.2. Find P(exactly two late).

(d) [5 pt] Preparation time T ~ Normal(50, 8^2) minutes. Find P(T>62) and the 90th percentile.

Answer Key & Detailed Solutions

정답 확인은 문제를 먼저 끝까지 풀고 진행하세요. 각 해설에는 공식과 계산의 근거를 포함했습니다.

PART A · ANSWERS & EXPLANATIONS

MC1 C MC2 C MC3 C MC4 B MC5 B MC6 C MC7 C MC8 A MC9 A MC10 B

MC1 · C 여러 농가라는 횡단면 단위와 여러 연도라는 시간축이 동시에 있으므로 패널자료다.

MC2 · C 1-1/k²에서 k=2이면 1-1/4=0.75. 이는 정규분포 여부와 무관하다.

MC3 · C P(A∩B)=0.5×0.4=0.2, 따라서 P(A∪B)=0.5+0.4-0.2=0.7.

MC4 · B P(X=1)=e^(-3)×3¹/1!≈3×0.0498=0.1494.

MC5 · B (20-12)/(36-12)=8/24=1/3.

MC6 · C n=7이므로 L75=0.75×8=6. 여섯 번째 관측값은 60.

MC7 · C 이항분포 기댓값 E(X)=np=10×0.2=2.

MC8 · A -0.9는 강한 음의 선형관계를 의미한다. 인과관계나 독립성을 직접 결론 낼 수 없다.

MC9 · A Z=(90-80)/5=2, 따라서 P(Z>2)=1-Φ(2)=0.0228.

MC10 · B 독립일 때 Var(R-C)=3²+4²=25. 표준편차는 √25=5이다. 분산과 표준편차를 혼동하지 않는다.

PART B · ANSWERS & EXPLANATIONS

TF1 · TRUE 적절한 무작위 표본추출에서 E(x-bar)=μ이다. 다만 특정 한 표본의 평균이 반드시 μ와 같다는 뜻은 아니다.

TF2 · FALSE 양의 확률인 상호배반사건의 교집합은 0이므로 독립사건 곱셈식과 충돌한다.

TF3 · FALSE 조건부확률에서 분모가 각각 P(B), P(A)로 달라 일반적으로 값도 다르다.

TF4 · TRUE Var(X)=E(X²)-[E(X)]²=21-16=5.

TF5 · FALSE 표준편차는 변하지 않지만 평균이 커지므로 CV=s/x-bar는 보통 감소한다.

TF6 · TRUE 독립성이 있으면 곱의 기댓값이 기댓값의 곱으로 분리된다(기댓값들이 존재할 때).

TF7 · TRUE PDF는 점의 확률이 아니라 확률밀도다. 전체 적분이 1이면 부분적으로 f(x)>1일 수 있다.

TF8 · FALSE 포아송분포는 0,1,2,…와 같은 사건의 발생 횟수를 모델링한다.

TF9 · TRUE P(T>s+t | T>s)=P(T>t), 즉 이미 기다린 시간에 영향을 받지 않는다.

TF10 · FALSE 이항분포는 Var(X)=np(1-p)이다. np는 평균에 해당한다.

PART C · STEP-BY-STEP SOLUTIONS

C1. Weekly market sales

(a) 합계는 200이므로 평균=200/8=25. 중앙값=(22+25)/2=23.5. 평균이 중앙값보다 약간 크며 오른쪽 꼬리의 가능성을 시사한다.

(b) L25=2.25→Q1=18+0.25(20-18)=18.5. L75=6.75→Q3=27+0.75(30-27)=29.25. IQR=10.75. 상한=29.25+1.5×10.75=45.375. 최댓값 43은 이보다 작으므로 1.5×IQR 기준의 상측 이상치는 없다.

(c) 평균 25를 기준으로 제곱편차합=100+49+25+9+0+4+25+324=536. s^2=536/7≈76.5714, s≈8.7505. CV=8.7505/25×100≈35.00%.

C2. Two crop-pest tests

(a) P(+)=0.12×0.85+0.88×0.10=0.102+0.088=0.190.

(b) P(D|+)=0.102/0.190≈0.5368. 양성이라는 정보가 감염 확률을 12%에서 약 53.7%로 높인다.

(c) 감염 여부를 조건으로 두 검사 결과가 독립이므로 P(+,+|D)=0.85^2, P(+,+|D^c)=0.10^2. 따라서 P(D|+,+)=(0.12×0.85^2)/(0.12×0.85^2+0.88×0.10^2)=0.0867/0.0955≈0.9079.

C3. Delivery orders, profit, and preparation

(a) E(X)=0.35+2×0.35+3×0.15=1.5. E(X^2)=0.35+4×0.35+9×0.15=3.10. Var(X)=3.10-2.25=0.85, sd(X)=√0.85≈0.9220.

(b) E(Y)=30×1.5-12=33(천 원). sd(Y)=30√0.85≈27.659(천 원). 비용 -12는 표준편차에 영향을 주지 않는다.

(c) K~Binomial(4,0.2). P(K=2)=C(4,2)×0.2^2×0.8^2=6×0.04×0.64=0.1536.

(d) Z=(62-50)/8=1.5 → P(T>62)=1-Φ(1.5)=1-0.9332=0.0668. 90백분위수=50+1.2816×8≈60.2528분.

FINAL FORMULA & ERROR CHECK

Descriptive	Sample variance s^2=Σ(xi-x-bar)^2/(n-1); sd=√s^2; CV=sd/x-bar ×100%; IQR=Q3-Q1.
Probability	P(A ∪ B)=P(A)+P(B)-P(A ∩ B); P(A|B)=P(A ∩ B)/P(B).
Bayes	P(D|+)=P(+|D)P(D)/[P(+|D)P(D)+P(+|D^c)P(D^c)].
Moments	E(X)=Σx p(x); Var(X)=E(X^2)-[E(X)]^2; E(a+bX)=a+bE(X); Var(a+bX)=b^2Var(X).
Distributions	Binomial: E=np, Var=np(1-p). Poisson: P(X=k)=exp(-λ) × λ^k/k!, E=Var=λ.
Continuous	Uniform(a,b): P(c<X<d)=(d-c)/(b-a). Exponential(rate λ): P(T>t)=exp(-λt).

마지막 점검: ① 확률의 합 1 확인 ② 조건부확률의 분모 재확인 ③ n / n-1 구별 ④ E(X^2)와 [E(X)]^2 구별 ⑤ 분산과 표준편차의 단위 구별 ⑥ 독립성 가정 확인